

Tanzanian Water Wells: Predictive Maintenance for Clean Water Access

Using Data to Keep Safe Water Flowing

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46%

Of wells are not fully functional

59,400

Water points analyzed in this dataset

The Challenge

Why predictive maintenance is critical

①

Reactive Maintenance

Repairs typically happen only after a breakdown occurs, leaving communities without water for days or weeks.

②

Distance & Burden

When a local well fails, women and children must walk significantly further to alternative, often unsafe, sources.

③

Health & Safety Risks

Extended downtime forces communities to rely on surface water, drastically increasing the risk of waterborne diseases.

Objective

To leverage machine learning to predict water well functionality, shifting the paradigm from reactive repairs to proactive maintenance.

Ensure Reliability

Keep more wells functional for more days per year, ensuring communities have uninterrupted access to safe drinking water.

Optimize Resources

Help District Water Offices optimize repair routes, manage spare part inventory, and allocate budget where it's needed most.

Guide Investment

Provide data-driven insights to the Ministry of Water and NGOs to support strategic infrastructure planning across regions.

Who Benefits

Connecting stakeholders for sustainable water access through data-driven decisions

Ministry of Water

Enhancing national planning and data-driven policy making for long-term sustainability.

District Water Offices

Optimizing maintenance schedules, routing, and budget allocation for repair teams.

NGOs & Implementers

Improving targeted interventions and generating transparent impact reports for donors.

Local Communities

Ensuring reliable, safe, and nearby water access to support health and economic growth.

Understanding the Problem

The Ripple Effect of Well Failure

The Breakdown

A mechanical failure stops water flow. Without data, this breakdown often goes unreported for weeks.



The Burden

Women and children are forced to walk over 5km to alternative sources, carrying heavy loads daily.



The Consequence

School attendance drops, economic activity slows, and reliance on unsafe water increases disease risk.

Key Insight: Predicting failure isn't just about fixing pumps-it's about protecting health, education, and economic stability.



59,400

Training Wells

Labeled historical data used to teach the model

14,850

Test Wells

Unseen data used to validate performance



Key Features Analyzed:

Location (GPS), Water Quantity, Waterpoint Type, Construction Year, Management

Target Variable: Well Status

Functional

The pump is working and providing water. No immediate action needed.



Functional - Needs Repair

Still providing water but showing signs of wear. Critical for proactive maintenance.

Non-Functional

The pump has failed completely. Requires significant repair or replacement.



Paucek and Lage

Problem Statement



Consumers Demand Safe

Many bottled water brands face concerns over purity, sustainability, and taste consistency, leaving consumers seeking trustworthy alternatives.

Methodology

The Solution: Decision Tree

A Decision Tree model was chosen because it thinks like a human expert: asking a sequence of "Yes" or "No" questions to arrive at a diagnosis.

Transparent Logic

No "black box." We can trace exactly why a well is flagged as at-risk.

Actionable Rules

The model's logic translates easily into field guides for maintenance teams.

High Performance

Best performing model with validated accuracy on unseen data.



Purity & Safety

Tested and certified for quality and safety – reliable, pure, trusted, assured, consistent.



Sustainable Packaging

Eco-conscious bottles and refillable options – sustainable planet-friendly and innovative.



Hydration for Everyone

Suitable for individual, family, and corporate use – versatile, practical, convenient, inclusive.

Results That Matter

Translating Model Performance into Community Impact

75.88 %

Prediction Accuracy

Our Tuned Decision Tree model correctly identifies the operational status of 3 out of every 4 wells, significantly outperforming random chance and baseline predictions.

74.29%

Weighted F1-Score:

Balanced performance metric ensuring we don't just predict the majority class but accurately identify broken wells too.

Fewer Missed Failures

The Win: We catch non-functional wells before communities suffer long downtimes, enabling proactive repair scheduling.

Reduced False Alarms

The Savings: We avoid sending expensive repair crews to wells that are actually working fine, saving budget for where it's needed.

Benefits for Customers

Clean, refreshing hydration that's convenient, eco-friendly, and perfect for any lifestyle.



Health & Safety

Reliable, clean hydration – refreshing, pure, healthy, revitalizing, essential.



Eco-Friendly Choice

Sustainable packaging reduces environmental impact – eco-friendly, responsible and green.



Convenience

Easy access at home or on-the-go – convenient, versatile and practical, effortless.

Paucek and Lage

www.reallygreatsite.com

Case Example



Corporate Wellness Program

Corporate partnership provided healthy hydration, boosted employee satisfaction, and promoted eco-friendly practices.





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Why Paucek and Lage

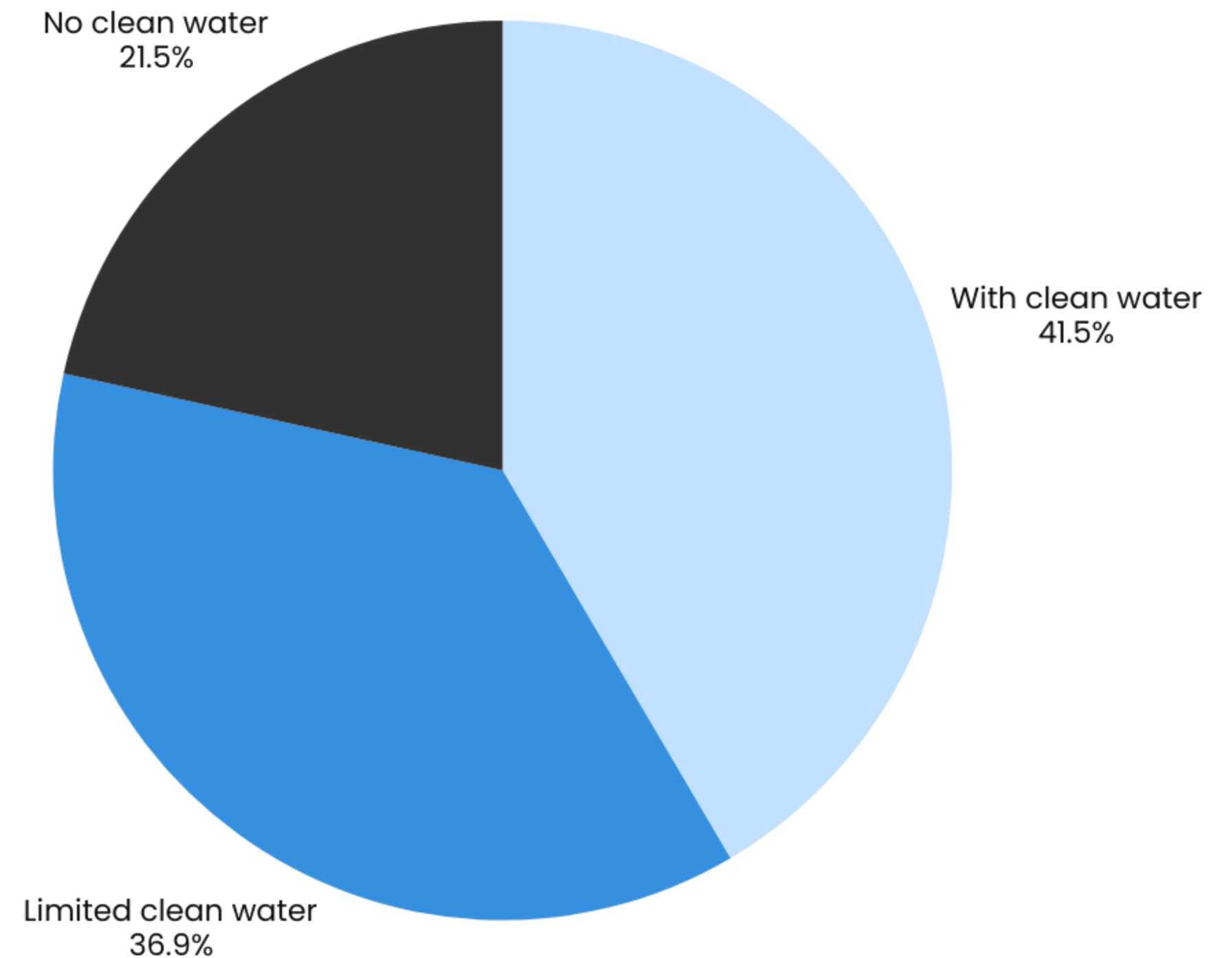
Sustainable, and Trusted Hydration

Expert mineral water production with pure, sustainable quality and flexible solutions, backed by a passionate, trusted team.

Local regions in need of clean water

Our data shows that within the San Gabriel community, the areas with clean water are steadily increasing.

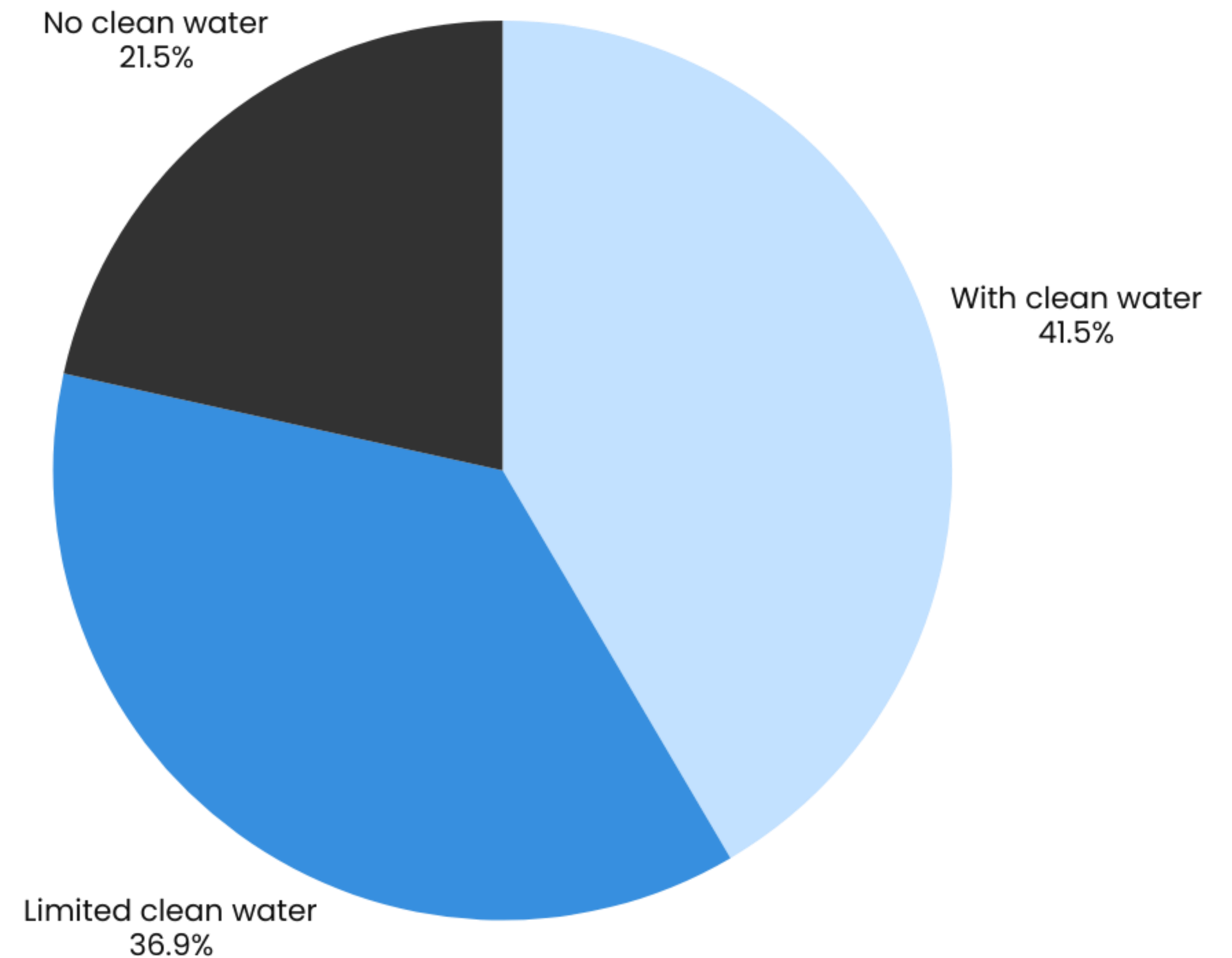
Paragraph that explains the data sources, outlines what the chart illustrates, and provides guidance on interpreting the visualizations to inform decision-making.



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Still, our goal is to ensure that over 70% of people will soon have reliable access to safe water in the **San Gabriel Community.**